Online Examination Report

Date:	28.07.2014
Attempts:	1
Examinee:	Kibuuka, Elvis
Date of birth:	11/16/1989
Subject:	020-AGK
Result:	0 %
Time used:	00:00:15 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	843	21.2.4	Fuselage, landing gear, doors, floor, etc	0,00	1,00
2	220	21.2.4	Fuselage, landing gear, doors, floor, etc	0,00	1,00
3	2358	21.3.1	Hydro-mechanics: basic principles	0,00	1,00
4	670	21.3.1	Hydro-mechanics: basic principles	0,00	1,00
5	2361	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
6	129	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
7	716	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
8	2297	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
9	2265	21.8.2.2	Design, operation, system components, etc	0,00	1,00
10	1763	21.8.2.2	Design, operation, system components, etc	0,00	1,00
11	2285	21.8.2.2	Design, operation, system components, etc	0,00	1,00
12	2250	21.9.1	General, definitions, basic applications: etc	0,00	1,00
13	2122	21.9.1.8	Semiconductors and logic circuits:	0,00	1,00
14	1735	21.9.3.1	DC Generation	0,00	1,00
15	942	21.10.1.2	Engine: design, operation etc	0,00	1,00
16	1353	21.10.1.2	Engine: design, operation etc	0,00	1,00
17	790	21.10.7	Ignition circuits	0,00	1,00
18	2470	21.10.10.1	Performance	0,00	1,00
19	308	21.11.1.2	Design, types of turbine engines, components	0,00	1,00
20	1131	21.13	OXYGEN SYSTEMS	0,00	1,00
21	3014	22.	INSTRUMENTATION	0,00	1,00
22	2797	22.	INSTRUMENTATION	0,00	1,00
23	3097	22.	INSTRUMENTATION	0,00	1,00
24	4130	22.	INSTRUMENTATION	0,00	1,00
25	2809	22.	INSTRUMENTATION	0,00	1,00
26	2815	22.	INSTRUMENTATION	0,00	1,00
27	3090	22.	INSTRUMENTATION	0,00	1,00
28	2806	22.	INSTRUMENTATION	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	2801	22.	INSTRUMENTATION	0,00	1,00
30	310	22.2.1	Pressure measurement	0,00	1,00
31	119	22.2.1	Pressure measurement	0,00	1,00
32	242	22.2.4	Altimeter	0,00	1,00
33	1024	22.2.4	Altimeter	0,00	1,00
34	1954	22.2.4	Altimeter	0,00	1,00
35	1045	22.2.4	Altimeter	0,00	1,00
36	1742	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	0,00	1,00
37	1755	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
38	489	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
39	1422	22.6.3	Flight Director : design and operation.	0,00	1,00
40	1655	22.12.3	Stall Warning Systems (SWS)	0,00	1,00
			Total:0%	**0,00**	**40,00**

Attachment 2: List of result by topic

Licence
Date: 28.07.2014
Examinee: Kibuuka, Elvis
Location: Uganda Civil Aviation Authority

Your Result: 0,00 from 40,00 Points (0,00 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
21.2. AIRFRAME	2	0,00	2,00	0,00
21.3. HYDRAULICS	2	0,00	2,00	0,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	4	0,00	4,00	0,00
21.8. FUEL SYSTEM	3	0,00	3,00	0,00
21.9. ELECTRICS	3	0,00	3,00	0,00
21.10. PISTON ENGINES	4	0,00	4,00	0,00
21.11. TURBINE ENGINES	1	0,00	1,00	0,00
21.13. OXYGEN SYSTEMS	1	0,00	1,00	0,00
22. INSTRUMENTATION	20	0,00	20,00	0,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	6	0,00	6,00	0,00
22.3. MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	1	0,00	1,00	0,00
22.4. GYROSCOPIC INSTRUMENTS	2	0,00	2,00	0,00
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	1	0,00	1,00	0,00
22.12. ALERTING SYSTEMS, PROXIMITY SYSTEMS	1	0,00	1,00	0,00
Total	**40**	**0,00**	**40,00**	**0,00**

1

With regard to an aircraft structure, 'fail safe' is one:

- ☐ *that is easily manufactured.*
- ☐ *that is only used for a limited time.*
- ☐ *in which the load is carried by other components if a part of the structure fails.*
- ☐ *used for small aircraft only.*

ID:	843(LTS:222426, LTM:182)	Max. Points	1,00
Syllabus reference:	21.2.4 Fuselage, landing gear, doors, floor, etc	Points	0,00

2

The fuselage of an aircraft consists, among others, of stringers whose purpose is to:

- ☐ **integrate the strains due to pressurization to which the skin is subjected and convert them into a tensile stress.**
- ☐ **assist the skin in absorbing the longitudinal traction-compression stresses.**
- ☐ **provide sound and thermal isolation.**
- ☐ **withstand the shear stresses.**

ID:	220(LTS:222290, LTM:46)	Max. Points	1,00
Syllabus reference:	21.2.4 Fuselage, landing gear, doors, floor, etc	Points	0,00

3

In a hydraulic system, the reservoir is pressurized in order to:

- ☐ **reduce fluid combustibility**
- ☐ **keep the hydraulic fluid at optimum temperature**
- ☐ **seal the system**
- ☐ **prevent pump cavitation**

ID:	2358(LTS:222837, LTM:963)	Max. Points	1,00
Syllabus reference:	21.3.1 Hydro-mechanics: basic principles	Points	0,00

4

The function of a hydraulic selector valve is to:

- ☐ ***select the system to which the hydraulic pump should supply pressure.***
- ☐ ***discharge some hydraulic fluid if the system pressure is too high.***
- ☐ ***direct system pressure to either side of the piston of an actuator.***
- ☐ ***automatically activate the hydraulic system.***

ID:	670(LTS:222386, LTM:142)	Max. Points	1,00
Syllabus reference:	21.3.1 Hydro-mechanics: basic principles	Points	0,00

5

In the pneumatic supply system of a modern transport aircraft, the air pressure is regulated. This pressure regulation occurs just before the manifold by the :

- ☐ **intermediate pressure check-valve**
- ☐ **high pressure bleed air valve**
- ☐ **low pressure bleed air valve**
- ☐ **fan bleed air valve**

ID:	2361(LTS:222839, LTM:965)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	0,00

6

On most modern airliners the cabin pressure is controlled by regulating the:

- ☐ **RPM of the engine.**
- ☐ **Bleed air valve.**
- ☐ **Airflow entering the cabin.**
- ☐ **Airflow leaving the cabin.**

ID:	129(LTS:222270, LTM:26)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	0,00

7

The cabin rate of descent is:

- ☐ **is not possible at constant airplane altitudes.**
- ☐ **always the same as the airplane's rate of descent.**
- ☐ **a cabin pressure increase.**
- ☐ **a cabin pressure decrease.**

ID:	716(LTS:222399, LTM:155)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	0,00

8

Assuming cabin differential pressure has attained the required value in normal flight conditions, if flight altitude is maintained:

- ☐ **the pressurisation system ceases to function until leakage reduces the pressure.**
- ☐ **a constant mass air flow is permitted through the cabin.**
- ☐ **the outflow valves will move to the fully open position.**
- ☐ **the outflow valves will move to the fully closed position.**

ID:	2297(LTS:222803, LTM:940)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	0,00

9

On most transport aircraft, the low pressure pumps of the fuel system are:

- ☐ **Diaphragm pumps.**
- ☐ **Gear type pumps.**
- ☐ **Piston pumps.**
- ☐ **Centrifugal pumps.**

ID:	2265(LTS:222796, LTM:933)	Max. Points	1,00
Syllabus reference:	21.8.2.2 Design, operation, system components, etc	Points	0,00

10

The cross-feed fuel system enables:

- ☐ **only the transfer of fuel from the centre tank to the wing tanks.**
- ☐ **the supply of the outboard jet engines from any outboard**
- ☐ **the supply of the jet engines mounted on a wing from any fuel tank within that wing.**
- ☐ **the supply of any jet engine from any fuel tank.**

ID:	1763(LTS:222672, LTM:429)	Max. Points	1,00
Syllabus reference:	21.8.2.2 Design, operation, system components, etc	Points	0,00

11

On a transport type aircraft the fuel tank system is vented through:

- ☐ **Bleed air from the engines.**
- ☐ **Ram air scoops on the underside of the wing.**
- ☐ **A pressure regulator in the wing tip.**
- ☐ **The return lines of the fuel pumps.**

ID:	2285(LTS:222801, LTM:938)	Max. Points	1,00
Syllabus reference:	21.8.2.2 Design, operation, system components, etc	Points	0,00

12

A Constant Speed Drive aims at ensuring

- ☐ **that the electric generator produces a constant frequency.**
- ☐ **that the CSD remains at a constant RPM not withstanding the generator RPM withstanding the acceleration of the engine.**
- ☐ **equal AC voltage from all generators.**
- ☐ **that the starter-motor maintains a constant RPM not**

ID: 2250(LTS:222791, LTM:928)
Syllabus reference: 21.9.1 General, definitions, basic applications: etc

Max. Points 1,00
Points 0,00

13

When a conductor cuts the flux of a magnetic field :

- ☐ **current will flow in accordance with Flemings left hand rule.**
- ☐ **the field will collapse.**
- ☐ **there will be no effect on the conductor.**
- ☐ **an electromotive force (EMF) is induced in the conductor.**

ID:	2122(LTS:222757, LTM:894)	Max. Points	1,00
Syllabus reference:	21.9.1.8 Semiconductors and logic circuits:	Points	0,00

14

In order that DC generators will achieve equal load sharing when operating in parallel, it is necessary to ensure that :

- ☐ **the synchronising bus-bar is disconnected from the busbar system.**
- ☐ **adequate voltage differences exists.**
- ☐ **their voltages are almost equal.**
- ☐ **equal loads are connected to each generator busbar before paralleling.**

ID: 1735(LTS:222631, LTM:418)
Syllabus reference: 21.9.3.1 DC Generation

Max. Points 1,00
Points 0,00

15

On modern carburettors, the variations of mixture ratios are obtained by the adjustment of :

- ☐ **fuel flow.**
- ☐ **fuel flow, air flow and temperature.**
- ☐ **fuel flow and air flow.**
- ☐ **air flow.**

ID: 942(LTS:222457, LTM:203)
Syllabus reference: 21.10.1.2 Engine: design, operation etc

Max. Points 1,00
Points 0,00

16

The power of a piston engine which will be measured by using a friction brake is :

- ☐ **Heat loss power.**
- ☐ **Brake horse power.**
- ☐ **Friction horse power.**
- ☐ **Indicated horse power.**

ID:	1353(LTS:222553, LTM:309)	Max. Points	1,00
Syllabus reference:	21.10.1.2 Engine: design, operation etc	Points	0,00

17

If the ground wire between the magnetos and the ignition switch becomes disconnected the most noticeable result will be that:

- ☐ a still operating engine will run down
- ☐ the power developed by the engine will be strongly reduced
- ☐ the engine cannot be shut down by turning the ignition switch to the "OFF" position
- ☐ the engine cannot be started with the ignition switch in the "ON" position

ID:	790(LTS:222413, LTM:169)	Max. Points	1,00
Syllabus reference:	21.10.7 Ignition circuits	Points	0,00

18

How does V(NE) speed vary with altitude?

☐ **Remains the same at all altitudes.**

☐ **Varies directly with altitude.**

☐ **Varies inversely with altitude.**

ID: 2470(LTS:222912, LTM:1049)
Syllabus reference: 21.10.10.1 Performance

Max. Points 1,00
Points 0,00

19

The bypass ratio:
I. is the ratio of bypass air mass flow to HP compressor mass flow.
II. can be determined from the inlet air mass flow and the HP compressor mass flow.

- ☐ ***I is correct, II is incorrect.***
- ☐ ***I is incorrect, II is incorrect.***
- ☐ ***I is incorrect, II is correct.***
- ☐ ***I is correct, II is correct.***

ID:	308(LTS:222309, LTM:65)	Max. Points	1,00
Syllabus reference:	21.11.1.2 Design, types of turbine engines, components	Points	0,00

20

Consider the flight deck oxygen supply system. The purpose of the oxygen regulator (as a function of demand and altitude) is to:
1. decrease oxygen pressure from 1800 PSI (in the bottles) down to about 50-75 PSI (low pressure system)
2. supply pure oxygen
3. supply diluted oxygen
4. supply oxygen at normal pressure
5. supply oxygen at emergency/positive pressure
6. trigger the continuous cabin altitude warning at 10000 ft cabin altitude
The combination regrouping all the correct statements is:

- ☐ **1, 2, 3, 4**
- ☐ **2, 3, 4, 5**
- ☐ **3, 4, 5, 6**
- ☐ **1, 3, 4, 6**

ID: 1131(LTS:222496, LTM:253)
Syllabus reference: 21.13 OXYGEN SYSTEMS
Max. Points 1,00
Points 0,00

21

What is the significance of the blue radial line on the airspeed indicator of a light multiengine airplane and when is it to be used? It indicates the

- ☐ **speed which will provide the maximum altitude gain in a given time when one engine is inoperative and should be used for climb and final approach during engine-out operations.**
- ☐ **speed which will provide the greatest height for a given distance of forward travel when one engine is inoperative and should be used for all climbs during engine-out operations.**
- ☐ **minimum speed at which the airplane is controllable when the critical engine is suddenly made inoperative and should be used at all altitudes when an engine is inoperative.**

ID:	3014(LTS:222988, LTM:1125)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

22

Deviation error of the magnetic compass is caused by

- ☐ northerly turning error.
- ☐ the difference in location of true north and magnetic north.
- ☐ certain metals and electrical systems within the aircraft.

ID:	2797(LTS:222966, LTM:1092)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

23

How can a pilot identify a lighted heliport at night?

- ☐ **White and red beacon light with dual flash of the white.**
- ☐ **Green and white beacon light with dual flash of the white.**
- ☐ **Green, yellow, and white beacon light.**

ID:	3097(LTS:222993, LTM:1130)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

24

(Refer to Figure 85.) Of the following, which is the minimum acceptable rate of climb (feet per minute) to 9,000 feet required for the WASH2 WAGGE departure at a GS of 150 knots?

See Attachments:

1. instrument_85.jpg	figure 85	(Attachment No.1)

- ☐ **825 feet per minute.**
- ☐ **750 feet per minute.**
- ☐ **1,000 feet per minute.**

ID:	4130(LTS:223037, LTM:1174)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

25

Which instrument would be affected by excessively low pressure in the airplane's vacuum system?

- ☐ **Pressure altimeter.**
- ☐ **Airspeed indicator.**
- ☐ **Heading indicator.**

ID:	2809(LTS:222970, LTM:1096)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

26

If a pitot tube is clogged, which instrument would be affected?

- ☐ **Altimeter.**
- ☐ **Vertical speed indicator.**
- ☐ **Airspeed indicator.**

ID: 2815(LTS:222971, LTM:1097)
Syllabus reference: 22. INSTRUMENTATION

Max. Points 1,00
Points 0,00

27

Identify taxi leadoff lights associated with the centerline lighting system.

- ☐ **Alternate green and yellow lights curving from the centerline of the runway to the centerline of the taxiway.**
- ☐ **Alternate green and yellow lights curving from the centerline of the runway to a point on the exit.**
- ☐ **Alternate green and yellow lights curving from the centerline of the runway to the edge of the taxiway.**

ID:	3090(LTS:222991, LTM:1128)
Syllabus reference:	22. INSTRUMENTATION
Max. Points	1,00
Points	0,00

28

Which statement is true about magnetic deviation of a compass?

- ☐ **Deviation is the same for all aircraft in the same locality.**
- ☐ **Deviation is different in a given aircraft in different localities.**
- ☐ **Deviation varies for different headings of the same aircraft.**

ID:	2806(LTS:222969, LTM:1095)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

29

What should be the indication on the magnetic compass as you roll into a standard rate turn to the right from a south heading in the Northern Hemisphere?

- ☐ **The compass will indicate a turn to the right, but at a faster rate than is actually occurring.**
- ☐ **The compass will remain on south for a short time, then gradually catch up to the magnetic heading of the airplane.**
- ☐ **The compass will initially indicate a turn to the left.**

ID:	2801(LTS:222968, LTM:1094)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

30

Pressure Altitude at an airfield is indicated by an altimeter when the barometric sub-scale is set to:

- ☐ 1013.25 hPa
- ☐ QFE
- ☐ QNH

ID:	310(LTS:222310, LTM:66)	Max. Points	1,00
Syllabus reference:	22.2.1 Pressure measurement	Points	0,00

31

An aircraft is flying from a cold air mass into a warm air mass. The TAS and true altitude will:

- ☐ **TAS increases, true altitude decreases.**
- ☐ **Both increases.**
- ☐ **TAS decreases, true altitude increases.**

ID: 119(LTS:222269, LTM:25)
Syllabus reference: 22.2.1 Pressure measurement

Max. Points 1,00
Points 0,00

32

Transition altitude is obtained from:

- ☐ **ATC or VOR ATIS.**
- ☐ **1500 feet above airfield elevation.**
- ☐ **Jeppesen or Aerad flight guides.**

ID:	242(LTS:222296, LTM:52)	Max. Points	1,00
Syllabus reference:	22.2.4 Altimeter	Points	0,00

33

QNH is the sub-scale setting which will cause the altimeter to read:

- ☐ **vertical displacement above mean sea level**
- ☐ **the altitude at which any given pressure exists in the standard atmosphere**
- ☐ **pressure altitude**
- ☐ **vertical displacement above the airfield pressure datum**

ID:	1024(LTS:222450, LTM:226)	Max. Points	1,00
Syllabus reference:	22.2.4 Altimeter	Points	0,00

34

When ambient temperature is warmer than standard at a particular altitude, the altimeter will indicate:

- ☐ **lower than true altitude;**
- ☐ **the same as true altitude.**
- ☐ **higher than true altitude;**

ID:	1954(LTS:222693, LTM:856)	Max. Points	1,00
Syllabus reference:	22.2.4 Altimeter	Points	0,00

35

If the static vent became blocked during a descent, the altimeter would:

- ☐ read zero altitude
- ☐ under-read the correct altitude
- ☐ act normally
- ☐ over-read the correct altitude

ID:	1045(LTS:222477, LTM:233)	Max. Points	1,00
Syllabus reference:	22.2.4 Altimeter	Points	0,00

36

During a compass swing the following reading were noted: MAGNETIC HEADING(0-89-178-269), COMPASS HEADING(358-92-182-268). After correction for coefficients B and C, the compass reading on the westerly heading was:

- ☐ **266**
- ☐ **271**
- ☐ **270**

37

When, in flight, the needle of a needle-and-ball indicator is on the left and the ball on the right, the aeroplane is:

- ☐ *turning left with not enough left rudder pedal.*
- ☐ *turning right with not enough left rudder pedal.*
- ☐ *turning right with too much left rudder pedal.*
- ☐ *turning left with too much left rudder pedal.*

ID:	1755(LTS:222669, LTM:426)	Max. Points	1,00
Syllabus reference:	22.4.2 Rate of turn and slip indicator/Turn Co-ordinator	Points	0,00

38

When, in flight, the needle of a needle-and-ball indicator is on the right and the ball on the left, the aeroplane is:

- [ ] ***turning left with not enough right rudder pedal.***
- [ ] ***turning right with too much right rudder pedal.***
- [ ] ***turning left with too much right rudder pedal.***
- [ ] ***turning right with not enough right rudder pedal.***

ID:	489(LTS:222350, LTM:106)	Max. Points	1,00
Syllabus reference:	22.4.2 Rate of turn and slip indicator/Turn Co-ordinator	Points	0,00

39

The horizontal command bar of a flight director:

1 - repeats the position information given by the ILS in the horizontal plane
2 - repeats the position information given by the ILS in the vertical plane
3 - gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.

The combination that regroups all of the correct statements is:

- ☐ ***2.***
- ☐ ***1, 3.***
- ☐ ***3.***
- ☐ ***2, 3.***

ID:	1422(LTS:222571, LTM:328)	Max. Points	1,00
Syllabus reference:	22.6.3 Flight Director : design and operation.	Points	0,00

40

A stall warning system is based on measuring the:

- ☐ *TAS.*
- ☐ *groundspeed.*
- ☐ *angle of attack.*
- ☐ *attitude.*

ID: 1655(LTS:222646, LTM:393)
Syllabus reference: 22.12.3 Stall Warning Systems (SWS)

Max. Points 1,00
Points 0,00

Attachment No. 1

Questions 24

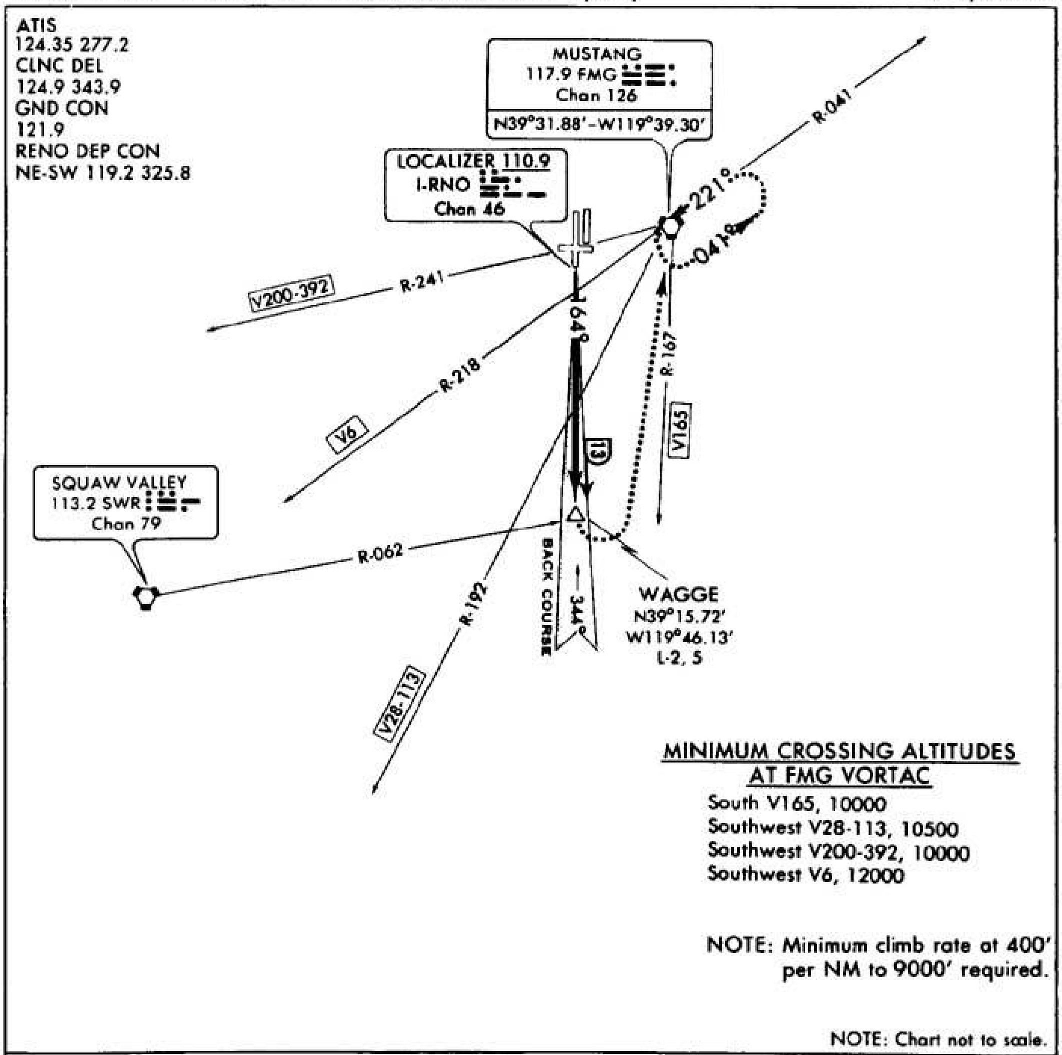
figure 85

instrument_85.jpg

(VECTOR) (WASH2.WAGGE) 91094

WASHOE TWO DEPARTURE SL-346 (FAA)

RENO CANNON INTL
RENO, NEVADA



DEPARTURE ROUTE DESCRIPTION

TAKE-OFF RUNWAYS 16L/R: Climb via I-RNO Localizer south course to WAGGE INT then via radar vectors to assigned route.

LOST COMMUNICATIONS: If not in contact with departure control within one minute after takeoff, or if communications are lost before reaching 9000', continue climb via I-RNO localizer south course to WAGGE INT, turn left, proceed direct FMG VORTAC. Cross FMG VORTAC at or above MCA, thence via assigned route or climb in holding pattern northeast on FMG R-041, left turns to cross FMG VORTAC at or above MCA for assigned route.

WASHOE TWO DEPARTURE
(VECTOR) (WASH2.WAGGE)

RENO, NEVADA
RENO CANNON INTL

Figure 85. WASHOE TWO DEPARTURE © ASA